



AR 600 Series

Multi-Component Analysers for Continuous Emissions Monitoring and Process Control

The Opsis AR 600 Analyser is the central unit of an Opsis Continuous Emissions Monitoring system. It accepts light from one or more lightpaths via a fibre optic cable and provides data for presentation through Opsis software. It also includes a modem, allowing data to be uploaded to a remote PC and allowing remote supervision and adjustment. The analyser has a high-performance spectrometer. Here, received light is converted into digital signals and analysed by the built-in computer. The computer software includes the spectrographic 'fingerprints' of a number of user-specified compounds. Using Beer Lambert's Law, it detects and measures these compounds through the depletions they cause in the spectrum of received light.

The system refines results by calculating margins of error for each value and by noting the proportion of transmitted light received by the analyser. This allows both data and system verification.

Opsis is a fully modular system which may be updated or expanded at any time. An Opsis analyser includes interfaces for a multiplexer (required when using multiple lightpaths) and for a datalogging system (required when integrating external signals from sensors) as standard. Furthermore, it is prepared for the addition of automatic calibration equipment. It is also prepared for purge air.

Updating an Opsis system to monitor additional gases often involves a simple software upgrade.

NO
NH₃

1, 2, 4-, 1, 3, 5-TMB
toluene, benzene
xylene, phenol
styrene, Hg
formaldehyde,
Cl₂, ClO₂, CS₂
O₃, SO₂, NO₂ and others

HF, H₂O
CH₄, HCl
CO, CO₂
N₂O
and
others

UV

IR

AR 600 series

AR 620 series

Wavelength ranges AR 600 series

AR 610 series

AR 650 series

Technical Specifications (standard)

Dimensions (L × W × H)	600 × 440 × 266 mm
Weight incl. case (approx.)	50 kg
Voltage supply	230 V (+ 6 %, - 10 %) / 115 V (± 10 %) 50/60 Hz
Power consumption	110 W
Computer	Embedded PC with VGA screen
Hard disk memory	2 Gb or more
Floppy disk drive	3 1/2", 1.44 Mb
Modem	Hayes compatible
Serial output	RS 232C
Ambient temperature	+ 15° C to + 25° C (+ 60° F to + 80° F)
Degree of protection	IP 20

Standard separately ordered

Gas calibrations (see application sheets)

Specifications subject to change without notice

Options

AR 601 with one UV detector
AR 602 with two UV detectors
AR 603 with three UV detectors
AR 610 with one IR detector
AR 620 with one IR detector and one UV detector
AR 650 with one IR detector
UV filter 230 series
Floppy disk drive 120 Mb

Accessories

Temperature controlled cabinet
EnviMan ComVisioner, Reporter software
Calibration equipment
IO 256 Signal handling system
Ambient pressure and temperature sensors
Short-haul modems

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